

Ananya Penumudi

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Professional Summary

B.Tech (CSE–IoT) student with experience in embedded systems, AI, industrial automation, and systems engineering research. Completed industrial training at BHEL Bengaluru and authored a thesis on the KAVACH Automatic Train Protection System. Experienced in technical documentation, system architecture, data analysis, cybersecurity, and technology-driven solutions.

Education

VNRVJIET

B.Tech (CSE–IoT)

2023–Ongoing

- CGPA: 8.92/10 (Till VI Semester) — JNTUH affiliated

Sri Chaitanya Junior College

Intermediate

2021–2023

- GPA: 97.4/100 — Telangana State Board

Bharatiya Vidya Bhavans Public School

High School

2021

- GPA: 9.38/10 — CBSE Board

Experience

Industrial Trainee

Jun 2026–Jul 2026

Bharat Heavy Electricals Limited (BHEL), Bengaluru

- Completed industrial training at BHEL Solar Business Division, gaining exposure to industrial manufacturing processes, production workflows, and large-scale engineering systems.
- Studied railway safety technologies, signalling infrastructure, and KAVACH-related concepts through technical discussions and documentation.
- Observed quality assurance, system integration, industrial project lifecycle management, safety-critical systems, and engineering operations.

Software Engineering Intern – Position Tracker Project

Jan 2026

Alfago

- Contributed to the development of a Position Tracker system for monitoring transaction-level and portfolio-level profit and loss (PnL).
- Implemented tax-lot based PnL calculations to support accurate financial reporting and portfolio analysis.
- Collaborated on backend business logic for transaction processing and portfolio performance tracking.

Operations Intern

Apr 2025–Jul 2025

Kartgen Infotech LLP

- Supported business development and operational activities during a three-month remote internship.
- Assisted in process coordination, stakeholder communication, and organizational workflow management.
- Worked under the guidance of Shrivatsa Malviya.

Projects

Smart AI Wheelchair System

Jan 2026

IoT + Edge AI

- Developed an AI-powered smart wheelchair with vision-based obstacle detection and automatic braking for collision prevention.
- Implemented GPS geofencing, caregiver alerts, and sensor fusion using camera, IMU, and proximity sensors.

Beyond Collision Prevention: A Systems Engineering Analysis of KAVACH

Jun 2026

Research Thesis

- Conducted a systems engineering analysis of KAVACH, India's indigenous Automatic Train Protection (ATP)

system.

- Studied system architecture, operational modes, communication framework, and functional safety mechanisms, and prepared technical documentation under industry guidance from BHEL Bengaluru.

AI-Enhanced Secure Data Wiping Platform

Nov 2025

AI + Cybersecurity

- Built a secure data sanitization platform with cryptographic verification and AI-based anomaly detection.

Auralyn – AI-Powered Emotional Wellness System

2025–Ongoing

IoT + AI + Healthcare

- Developing a wearable emotional wellness ecosystem integrating biometric sensing with AI-driven mood analysis.
- Built on ESP32-S3 with BLE connectivity for patient, therapist, and family-facing applications.

Syncope – Fall Detection & Alert System

Aug 2025

Wearable IoT

- Developed a wearable fall-detection system using motion sensors with automated caregiver alert mechanisms.

FocusFlow – ADHD-Friendly Productivity Platform

Apr 2025

AI + Web

- Built a productivity platform featuring mood tracking, journaling, facial-expression analysis, and gamified focus tools.

Ideavoid – Startup Idea Evaluation Platform

Nov 2024

- Built a platform for startup idea validation, peer feedback, and market viability assessment.

Technical Skills

Programming Languages: C, C++, Python, Java, JavaScript, HTML, CSS, R, DSA

Data & Analytics: Tableau, SQL, MySQL, Data Visualization, AI-based Anomaly Detection

AI & Computer Vision: Computer Vision, Edge Inference, Mood Detection

Systems Engineering: Requirements Analysis, Functional Safety, Verification & Validation, System Architecture, Requirements Traceability, Safety-Critical Systems

IoT & Embedded Systems: Arduino, ESP32, Sensor Integration (SPI, I2C, PWM), GPS, GSM, Accelerometer

Cloud & Databases: AWS, MySQL, SQL, MongoDB, Cloud-based Workflows

Cybersecurity: AI-based Anomaly Detection, Cryptographic Verification

Design & Tools: AutoCAD, StarUML, MS Office, MS Excel, Figma

Certifications

DevOps with Real-Time Practical Exposure

Jul–Nov 2025

Brainovision Solutions India Pvt. Ltd. – AICTE Approved – VNRVJIET

- Completed hands-on training in continuous integration, automation, containerization, and cloud-based DevOps workflows.

Tableau Certified Data Analyst Training

Jun 2024

Udemy – Instructor: Jed Guinto

- Completed a 59-hour certification course covering data visualization, dashboard design, and analytics using Tableau.

RoboJAM Workshop

Dec 2023

VNRVJIET – CONVERGENCE 2K23

- Participated in a hands-on robotics workshop focused on embedded systems and automation.

Achievements & Extracurricular Activities

- **2nd Prize in IDEATHON 2023**, VNRVJIET.
- Participated in **Smart India Hackathon (SIH) 2025** and contributed to solution design, prototyping, and technical documentation.
- Participated in **24-Hour Hackathon for Convergence 2025**, VNRVJIET.
- Participated in **24-Hour Hackathon DEMUX 2024**, BVRIT.
- Participated in **National Conclave 2025**, CBIT.
- Member, **Computer Society of India (CSI)**, VNRVJIET Student Chapter; active participant in organizing technical workshops, coding events, and inter-college competitions.
- Member, **Art Of Living (AOL)**, VNRVJIET Cultural Club; contributed to organizing college-level events and volunteering activities.
- **Languages:** English, Hindi, Telugu.